An Adaptive and Adjustable Maximum-Likelihood Estimator for SAR Change Detection

Mengmeng Wang, Jixian Zhang, Kazhong Deng, Huifu Zhuang, *Member, IEEE*, and Fenfen Hua

***Abstract*— Synthetic aperture radar (SAR) change detection (CD) can be broadly classified into two categories: noncoherent intensity CD and coherent change detection (CCD). The former methods, belonging to the field of image processing, are theoretically suitable for all SAR images since they only utilize the information of SAR magnitude; however, the detection precision cannot be guaranteed. The latter methods can show effective performances for most SAR image pairs using identical collection geometrics because of the basis of the probability and statistics theory. In this article, we propose a novel change estimator to combine the advantages of the two kinds of algorithms in a theoretical derivation way. An intensity-based estimator, inspired by the derivation of the coherence estimator in CCD, is first proposed. It is a new maximum-likelihood (ML) change estimator maximizing the probability distribution function (pdf) of the ratio change statistic instead of SAR complex data. In addition, the simple linear iterative cluster (SLIC) algorithm is introduced to make the new estimator adjustable because changes at different degrees can be extracted with varying settings of the superpixels number, which is further demonstrated in real SAR images. Finally, experiments in SAR image pairs of different statistical characteristics show that the proposed estimator can yield higher contrast SAR CD images than the other five common change statistics and obtain better CD maps than the other four classic thresholding methods.**

***Index Terms*— Change detection (CD), maximum-likelihood (ML) estimator, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) has been an important modality in remote sensing due to its ability to acquire high-resolution images with relative invariance to atmospheric, weather, and sunlight conditions [1]. As a hot application of SAR, SAR change detection (CD) utilizes two temporal SAR images of the same scene to identify changes that have occurred between the collection times [2].

Without regard to methods using machine learning, SAR CD algorithms can be broadly classified into two categories: noncoherent intensity CD using the SAR magnitude images to identify changes and coherent change detection (CCD) utilizing SAR phase and magnitude to estimate the coherence between the two images [3]. Although different in usage of image information, these two types of methods essentially have something in common. Specifically, the change statistics are employed to generate the corresponding images, such as various difference images (DIs) [4]–[8] for the intensity-based approaches and (complex) coherence images [3], [9]–[11] for CCD. After some postanalysis of these images, we can obtain the final change maps. Therefore, the selection of change statistics has an important influence on the results of CD.

Although needing to select change statistics, the theoretical bases on which the above two types of algorithms depend are different. Because generally utilizing the differences such as in intensity, texture, and spatial information [12]–[15] in the initial or transformed SAR images to identify changes, the intensity-based methods belong to the field of image processing. Therefore, the methods can be theoretically suitable for all SAR images to detect changes but not guarantee the detection precision due to the inevitable distortions caused by sensors, atmosphere, terrain, and other sources in SAR images. Differently, the existing CCD methods are often based on the assumption that SAR images are assumed to be collections of spatially uncorrelated pixels from a zero-mean circularly complex Gaussian distribution [16]. The key of CCD is to estimate the coherence; the sample degree of which is seen as a measure of the temporal change. Details of these algorithms will be described later in Section II-A. Because of the basis of the probability and statistics theory, the CCD algorithms can give effective performances for most SAR images of identical collection geometries.

To sum up, the intensity-based algorithms cannot guarantee the detection precision despite the applicability of all SAR image pairs, whereas CCD can give effective performances for most interferometric SAR images. To combine the advantages of these two types of algorithms and improve the detection precision, some works use both the intensity and coherence properties of SAR images to identify changes. In [14], the incoherent and coherent information is fused to generate DI by a ratio operator to get the final CD result. In [3], a two-stage method is presented to detect changes by using the noncoherent intensity method in the first stage and CCD in the

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second stage. These methods can be seen as the combination of the detection results of the two types of methods. Although improving the detection accuracy to a certain extent, they inevitably inherit their shortcomings.

Consequently, we take a different approach to propose a novel method that can combine the two types of algorithms in a theoretical derivation way. Similar to the derivation of the coherence estimators in CCD, we first derive a new intensity-based estimator, thus making it no longer limited to images with identical collection geometries. The focus of the derivation is getting the maximum-likelihood (ML) estimation of the coherence that maximizes the probability distribution function (pdf) of the ratio change statistic instead of SAR complex data. Moreover, we demonstrate that the estimator can be viewed as an adaptive thresholding algorithm. Second, the simple linear iterative cluster (SLIC) [17], [18] algorithm is introduced into the estimation process of the proposed estimator; the purpose of which is to identify changes at different degrees according to the task requirements, thus improving the applicability of the proposed method. The main advantages of the presented algorithms are listed as follows.

1) *Adaptive*: The threshold of a pixel can be automatically determined according to the ratio information in the homogenous region containing the pixel; then, the intensity of the pixel is positively or negatively stretched with reference to the threshold.
2) *Adjustable*: The changes at different degrees can be detected by setting the number of segmentation labels in SLIC.

The remainder of this article is organized as follows. In Section II, we derive a new change estimator inspired by the derivation of the coherence statistics in CCD, prove the essence of the estimator, and provide the introduction way of the SLIC algorithm into the estimation of the proposed estimator. Section III discusses the setting of superpixels number to extract changes at different degrees and then evaluates the new estimator by comparing it with commonly used intensity and coherent change statistics. Section IV concludes this article. To facilitate reading, Table I lists the symbols and acronyms used in this article.

TABLE I
LIST OF SYMBOLS AND ACRONYMS

Symbol	Description	First appearance
R	Ratio of the true variances	Fig. 1
$\boldsymbol{f}, \boldsymbol{g}$	Data sets of N pixels from two SAR images	Before Eq. (1)
$\boldsymbol{X}$	A joint data vector formed by $\boldsymbol{f}$ and $\boldsymbol{g}$	Before Eq. (1)
$\boldsymbol{Q}$	Covariance matrix	Eq. (1)
$\lvert\ \rvert$	Determinant of a matrix or modulo of a vector	Eq. (1)
H	Superscript for the complex conjugate transpose operation	Eq. (1)
E	Mathematic expectation operation	Eq. (2)
γ	Magnitude of a complex correlation coefficient	Eq. (2)
f, g	Subscripts for two SAR images	Eq. (2)
φ	Constant phase difference	Eq. (2)
σ^2	Variance	Eq. (3)
$\hat{}$	Estimator of one parameter	Eq. (3)
$\hat{R}$	Estimator of the true ratio	Eq. (4)
B	Beta function	Eq. (5)
R_c^0	Initial estimator defined in (8)	Fig. 3
$\hat{R}_m$	Mean ratio detector	Eq. (11)
I	Abbreviation for "intensity"	Eq. (11)
$\boldsymbol{\Omega}$	A data set of one segmented patch	Eq. (12)
k	Initial number of superpixels	Fig. 6

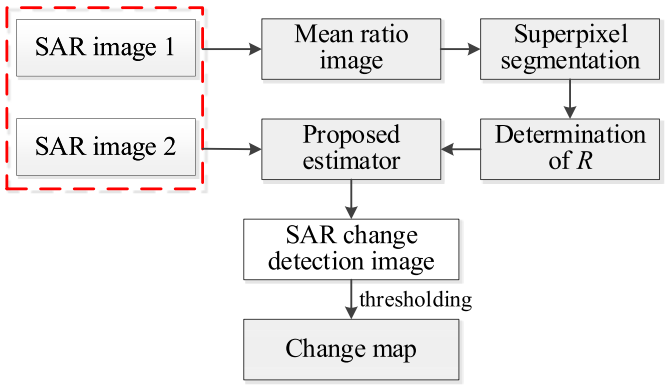


Fig. 1. Flowchart of the proposed CD method.

II. PROPOSED CD ALGORITHM

Fig. 1 shows the main steps of the proposed CD method. First, a mean-ratio (MR) image is generated using the similar operator in [5] from the spatial-registered SAR images and segmented using the SLIC algorithm. Then, according to the proposed method described in Section II-C, the parameter R in the new estimator is estimated based on the segmentation result. Finally, the SAR CD image is generated using the presented estimator, processing with the thresholding and some simple postclassification operations such as sieving and clumping to get the final CD map. Details of these steps are described in the following.

A. Proposed ML Estimator

Given two spatial-registered SAR images, one can form a data vector $\boldsymbol{X} = [\boldsymbol{f}, \boldsymbol{g}]^{\mathrm{T}}$ from the same N-pixel local neighborhood. The pairs of the vector $\boldsymbol{X_i} = [f_i, g_i]^{\mathrm{T}}, i = 1, 2, \ldots, N$ are often seen as the independent samples from a 2-D complex Gaussian distribution. And the pdf of the vector $\boldsymbol{X}$ is given by

$$p(\boldsymbol{X}) = \frac{1}{\pi^{2N}|\boldsymbol{Q}|^N} \exp\left(-\sum_{i=1}^{N} \boldsymbol{X_i^H}\, \boldsymbol{Q}^{-1} \boldsymbol{X_i}\right) \tag{1}$$

where $\boldsymbol{Q}$ is the covariance matrix given by

$$\boldsymbol{Q} = \mathbb{E}\left(\boldsymbol{X_i X_i^H}\right) = \begin{bmatrix} \sigma_f^2 & \gamma\, \sigma_f \sigma_g e^{j\varphi} \\ \gamma\, \sigma_f \sigma_g e^{-j\varphi} & \sigma_g^2 \end{bmatrix} \tag{2}$$

where $\sigma_f^2 = \mathbb{E}(|\boldsymbol{f}|^2)$, $\sigma_g^2 = \mathbb{E}(|\boldsymbol{g}|^2)$, γ denotes the magnitude of the complex correlation coefficient, and φ is the constant phase difference that exists between the two images in the local neighborhood.

Because of the speckles, SAR CD always uses the ratio approach, thus detecting change using the results of dividing one magnitude image by the other. The ML estimation of the mean backscatter power in the N-pixel local neighborhood can

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be obtained by averaging the intensity of all pixels in this area

$$\hat{\sigma}_f^2 = \frac{1}{N}\sum_{i=1}^{N}|f_i|^2$$
$$\hat{\sigma}_g^2 = \frac{1}{N}\sum_{i=1}^{N}|g_i|^2. \tag{3}$$

Therefore, the ratio at the corresponding location in $\boldsymbol{f}$ and $\boldsymbol{g}$ can be calculated using a spatial window and expressed as

$$\hat{R} = \frac{\hat{\sigma}_f^2}{\hat{\sigma}_g^2}. \tag{4}$$

The statistic $\hat{R}$ can be utilized well to test if the same location of two-temporal SAR images has been changed and its pdf is given by [19]

$$p(\hat{R}|R,\gamma,N) = \frac{\hat{R}^{N-1}(\hat{R}+R)R^N(1-\gamma^2)^N}{B(N,N)\left[(\hat{R}+R)^2-4\hat{R}R\gamma^2\right]^{N+\frac{1}{2}}}, \quad \hat{R}\geq 0 \tag{5}$$

where $B(N,N) = \Gamma^2(N)/\Gamma(2N)$ denotes the beta function, and $R = \sigma_f^2/\sigma_g^2$ is the ratio of the true variances.

The existing CCD algorithms often solve the ML estimate of the change statistics by maximizing (1) or its variants [20]–[24]. Due to relying on the statistical characteristics of SAR images, these statistics, such as the coherence in [9] and [10], can give a good performance for most two-pass repeat-geometry SAR image pairs. Initially, for a pair of interferometric SAR images, the coherence, the magnitude of the complex cross-correlation coefficient, is viewed as a measure of the degree of similarity to show the quality of interference fringes. Subsequently, it could be used as a change statistic to identify change areas in the scene. Inspired by the derivation of these coherence estimators, here, we take a different way to derive a new change measure by maximizing the pdf of $\hat{R}$ instead of $\boldsymbol{X}$ with the ML principle. The reasons are described as follows. As the MR statistic can effectively reflect the change information and weaken the speckle noise to some extent, intuitively, it seems more reasonable by maximizing (5) or, equivalently, the log of (5) instead of maximizing (1) to get the ML estimation of γ. To give the theoretical explanation, we plot the distributions of $\hat{R}$ for different coherence values of $\gamma = 0.0$, $\gamma = 0.3$, $\gamma = 0.6$, and $\gamma = 0.9$ when $R = 0.8$ and $N = 5$ (see Fig. 2). In Fig. 2, the red points indicate the max peaks of the corresponding pdf and the blue dotted line means that the estimation value $\hat{R}$ is equal to the true ratio value R. Note that the x-axis coordinate $\hat{R}$ of the peak point of the pdf gets closer to the ratio of the true variance with γ increasing. One point is worth mentioning that the peak point of pdf indicates that the probability of $\hat{R}$ equaling to the x-axis coordinate of this peak point is the largest.

In general, if a change has occurred between two samples taken from SAR images of the same scene, they would have a significant difference in intensity and a low similarity. Similarly, if no changes have occurred, the two samples would usually have similar intensities and a high similarity. Consider that the coherence value can be seen as a measure of the degree of similarity, there is a great consistency between the above description and the distributions of $\hat{R}$ with different coherence values. Specifically, in the unchanged case with R closing to 1.0 and a high coherence value, the probability of the estimation $\hat{R}$ closing to R will be high. In other words, $\hat{R}$ is very likely to be a large value close to R. In the change case with R away from 1.0 and a low coherence value, the probability of $\hat{R}$ away from R will be high. Here, we only consider the case of $R < 1$ when R is far from 1.0; the reason for which will be described later in Section II-B. At this point, $\hat{R}$ is very likely to be a small value much smaller than R. In short, if meeting the distribution as described in (5), the estimation $\hat{R}$ is close to the true ratio R (a value close to 1.0) in the unchanged case, but much less than R (a value far from 1.0) in the changed case, thus improving the contrast between the changes and no-changes. Inspired by this, we proposed a new ML estimator that maximizes the log of (5) concerning γ to make $\hat{R}$ meets the distribution with the greatest probability.

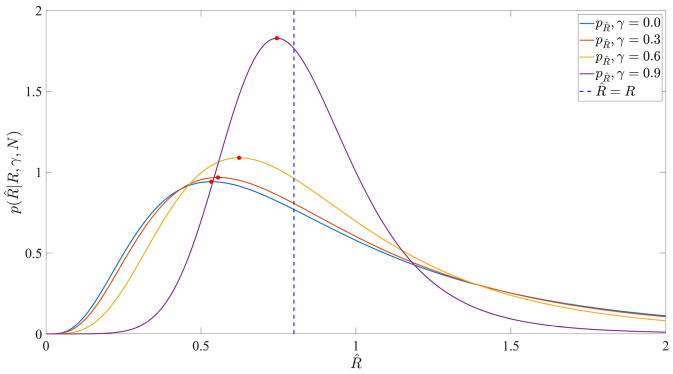


Fig. 2. Pdf $p(\hat{R}|R,\gamma,N)$ for $\gamma = 0.0$, 0.3, 0.6, and 0.9 when $R = 0.8$ and $N = 5$.

The log of (5) is easily got to be

$$\ln\left[p(\hat{R}|R,\gamma,N)\right] = \ln c + N\ln\left(1-\gamma^2\right) - \left(N+\frac{1}{2}\right) \cdot \ln\left[(\hat{R}+R)^2 - 4\hat{R}R\gamma^2\right] \tag{6}$$

where $c = \ln[\hat{R}^{N-1}(\hat{R}+R)R^N] - \ln B(N,N)$.

Taking the partial derivative of (6) with respect to γ and setting the derivative to zero gives

$$\frac{-N\gamma}{1-\gamma^2} + \frac{4(N+1/2)\hat{R}R\gamma}{(\hat{R}+R)^2-4\hat{R}R\gamma^2} = 0. \tag{7}$$

The available solution to (7) is

$$\gamma^2 = 1 - \frac{N(\hat{R}-R)^2}{2\hat{R}R}. \tag{8}$$

Here, the range values of γ^2 are first discussed. Since N, $\hat{R}$, and R are parameters greater than 0, it is easy to know $\gamma^2 \leq 1$. To get the lower limit of γ^2, we can get the following relationship after some algebraic manipulations:

$$\gamma^2 = (N+1) - \frac{N}{2}\left(\frac{\hat{R}}{R} + \frac{R}{\hat{R}}\right). \tag{9}$$

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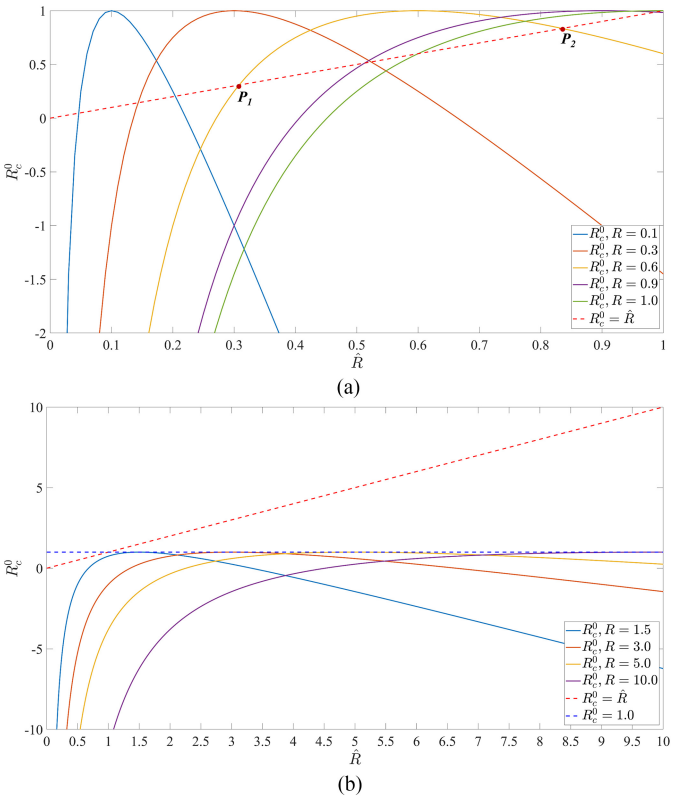


Fig. 3. R_c^0 versus $\hat{R}$ with different R values in the condition of: (a) $R \leq 1$ and (b) $R > 1$.

For a fixed value of R, it can be known that (9) tends to minus infinity as $\hat{R}$ tends to 0. Therefore, the value of (8) is within the interval $(-\infty, 1]$, which is the reason why we do not give the expression of γ. In this article, we choose the solution γ^2 as the new change estimator and mark it as R_c. As mentioned above, the assumption that the samples of the same local neighborhood are drawn independently of the same distribution should be met. But for some areas in SAR images, for example, the change areas where the intensities of the two samples often have significant differences, the assumption is less likely to be met. Therefore, the proposed estimator is slightly different from the typical coherence statistics and, according to the form of (8), more likely to fall into the category of ratio statistics.

In general, an optimal image generated by the change statistics should provide a contrast between the background (unchanged areas) and the changed regions as obvious as possible, i.e., restraining the information of the background and enhancing that of the changed areas to the greatest extent [12]. To achieve this goal, we define the final estimator as

$$R_c = \begin{cases} 1 - \dfrac{N(\hat{R} - R)^2}{2\hat{R}R}, & \text{if } \hat{R} \leq \hat{R}_P \\ \hat{R}, & \text{if } \hat{R} > \hat{R}_P \end{cases} \tag{10}$$

where $\hat{R}_P$ is the x-axis coordinate of the larger-value intersection point [see Fig. 3(a)] between the curve of (8) and the line $R_c = \hat{R}$. A detailed explanation of the definition will be given later in Section II-B.

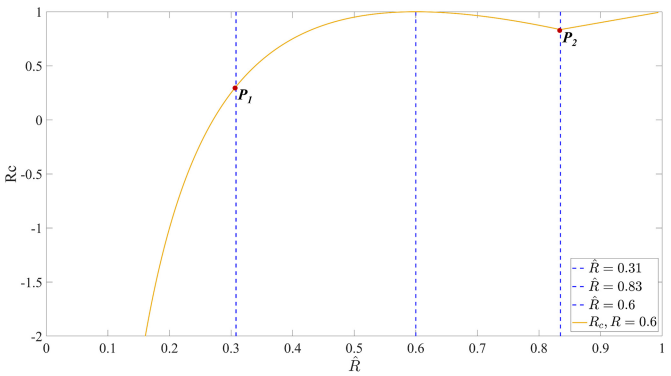


Fig. 4. Illustration of final R_c versus $\hat{R}$ when $R = 0.6$.

B. Essence of the Proposed Estimator

To explore the essence of the new estimator, Fig. 3 shows the distribution of (8) for different values of $\hat{R}$ when the sample number $N = 3$. In order to facilitate the explanation below, the estimator defined in (8) is referred to as R_c^0. Fig. 3(a) shows the curves of R_c^0 for the true ratio values of $R = 0.1$, $R = 0.3$, $R = 0.6$, $R = 0.9$, and $R = 1.0$ in the condition of $R \leq 1$, and Fig. 3(b) shows the curves for $R = 1.5$, $R = 3.0$, $R = 5.0$, and $R = 10.0$ in the condition of $R > 1.0$. The straight dotted line in red in Fig. 3 is auxiliary to display the relationship between the curve of (8) and the line $R_c^0 = \hat{R}$, favorable for explaining the definition of (10).

Note that there are two intersection points between the curve of (8) and the line $R_c^0 = \hat{R}$ in the condition of $R \leq 1$. Taking the curve of $R = 0.6$ as an example, the two intersections are marked as P_1 and P_2, respectively, and we observe that the estimator can stretch the statistic $\hat{R}$ to the decreasing direction in the condition of $\hat{R} < R_{P_1}$ or $\hat{R} > R_{P_2}$, whereas to the increasing direction in the condition of $R_{P_1} < \hat{R} < R_{P_2}$. Considering the fact that a good image generated from the change statistics should enhance the information of the change regions and restrain the information of the unchanged regions, we do not want the value of the estimator to decrease when $\hat{R} > R_{P_2}$. Therefore, we adopt the MR value as the value of the estimator in this condition, defining the final estimator as the form of (10). The orange curve in Fig. 4 intuitively shows the relationship between the final estimator R_c and $\hat{R}$ in the condition of $R = 0.6$.

Practically, the value range of R is $[0, +\infty)$. We have discussed the case of $R \leq 1$. In the case of $R > 1$, it can be seen from Fig. 3(b) that there are no intersection points any longer between the curve of (8) and the line $R_c = \hat{R}$, which means that the estimator R_c stretches the statistic $\hat{R}$ to the decreasing direction in this condition. This stretch effect is obviously no longer able to highlight the information of the change areas as that in the case of $R \leq 1$. To address this problem, the value range of R should be limited to [0, 1]. Fortunately, we can easily achieve this goal using the definition like (11). One point to be added here is that the dotted line in blue, i.e., $R_c = 1.0$, indicates the upper limit of the estimator R_c.

To intuitively display the stretching effect of the proposed estimator, we randomly simulated a ratio matrix of size

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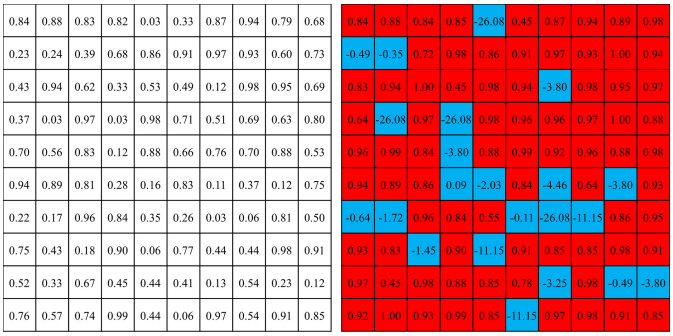


Fig. 5. Illustration of the stretching effect of the proposed estimator: (a) simulated ratio matrix and (b) stretched matrix where the elements marked in red are positively stretched and in blue are negatively stretched.

10×10 in the range [0, 1] [see Fig. 5(a)]. The corresponding stretched matrix obtained by the proposed estimator is shown in Fig. 5(b) where the elements marked in red are positively stretched and in blue are negatively stretched. In this test, the true ratios of all elements are assumed to be the same, that is, $R = 0.6$. Note that the proposed estimator can effectively stretch the small-value elements to the much smaller-value ones, thus significantly improving the contrast between the small- and the large-value elements. Here, the threshold of stretching can be easily solved by the two-degree equation, corresponding to the coordinate of point P_1.

Considering the above analysis, we can conclude that it is reasonable to regard the proposed estimator as an adaptive thresholding algorithm. For each pixel in SAR images, the threshold is different and can be automatically determined by the values of $\hat{R}$ and R.

Significantly, there is a doubt about the proportion of pixels with the estimation $\hat{R}$ larger than $\hat{R}_{\mathrm{P}}$ to the total pixels of the image. In other words, if too many pixels in SAR images meet the condition of $\hat{R} > R_{P_2}$, the proposed estimator would be meaningless. And the fact is that it has little influence on the detection results using values of the MR replacing that of (8) in the condition of $\hat{R} > R_{P_2}$. First, as the ML estimation of the true ratio, $\hat{R}$ should not be far from R in theory. For a specific example, in the condition of the true ratio $R = 0.6$, the probability that the value of the estimation $\hat{R}$ is in [0.31, 0.83] is relatively large and $\hat{R}$ is rarely possible larger than 0.83 (see Fig. 4). Here, the values 0.31 and 0.83 correspond to the coordinates of the intersection points P_1 and P_2, respectively. Moreover, we performed an experiment to illustrate the conclusion using the real SAR image pair, i.e., dataset 3 in Table II. The result shows that the proportion of the points with the estimation larger than R of the larger-value cross point is only 0.016% (1804 pixels in total), which further confirms our conclusion and the rationality of the definition.

C. Determination of R

It can be seen that the new change estimator defined in (10) requires the ratio of the true variances. However, it is almost impossible to get the real value of R. To address this problem, we can adopt alternatively the MR of a homogenous region as the true ratio for all pixels in this region. In 2003, Ren and Malik [25] first introduced the "superpixels" by oversegmenting an image in a preprocessing stage for image segmentation, which can group pixels into perceptually meaningful atomic regions with similar elements such as brightness and texture. Up to now, there are many methods for generating superpixels. Among them, the SLIC algorithm has become a very popular preprocessing step for image segmentation due to its simplicity of implementation and its outputs of better quality superpixels at a very low computational and memory cost [17]. Therefore, we choose the SLIC algorithm to segment SAR images and regard the segmented regions with similar elements as the homogenous area to estimate R.

TABLE II

AVAILABLE IMAGE PAIRS AND THEIR INFORMATION. TDX IS THE ABBREVIATION FOR TANDEM-X, AND TSX IS FOR TERRASAR-X

Set	Sensor	Resolution	Date	Multilook	Size
1	TDX	6m	16/11/2016	3×3	600×500
	TDX	6m	25/11/2017	3×3	600×500
2	TSX	6m	25/10/2015	3×3	2000×2000
	TSX	6m	21/01/2016	3×3	2000×2000
3	TDX	6m	21/11/2014	3×3	3333×3333
	TDX	6m	26/02/2016	3×3	3333×3333

The proposed procedure to generate the R image for an SAR image pair is as follows.

1) Generate an MR image from two spatial-registered SAR images using the following MR operator:

$$\hat{R}_m = \min\left\{\frac{\bar{I}_f}{\bar{I}_g}, \frac{\bar{I}_g}{\bar{I}_f}\right\} \tag{11}$$

where $\bar{I}_f$ and $\bar{I}_g$ stand for the local mean values of two SAR images, respectively. Then, segment the MR image using the SLIC algorithm.

2) For each segmented patch, average the MR of all pixels as the R value of this region

$$R = \frac{\sum_{\hat{R}_{m_i} \in \Omega} \hat{R}_{m_i}}{K}, \quad i = 1, 2, \ldots, K \tag{12}$$

where Ω is a set of one segmented patch extracted from the MR image, and the total pixel number of the patch is K, and $\hat{R}_{m_i}$ is the MR value of the ith pixel.

Taking a specific example, two real SAR images extracted from dataset 3 in Table II without the multilooking are utilized to illustrate the generation of the R image. Fig. 6(a) shows the two extracted images with the size of 800×800. The MR image is obtained using (11) with the window size 3×3 and then segmented with the SLIC algorithm. Fig. 6(b) shows the segmented results of the MR image with different desired initial numbers of superpixels, $k = 30$ and $k = 100$, respectively. One point to be mentioned here is that the segmented results correspond to images of overlaying superpixels maps on the MR image and the cyan lines are segment the boundaries of SLIC. The corresponding R images estimated by the proposed method are shown in Fig. 6(c). Each image block circled by the cyan line has the same R, that is, the mean value calculated in the way of step 2) within the corresponding block of the MR image. To intuitively display the way of generating R images, the images in Fig. 6(c) are also superimposed showings that the segmentation boundaries of SLIC overlay on R images.

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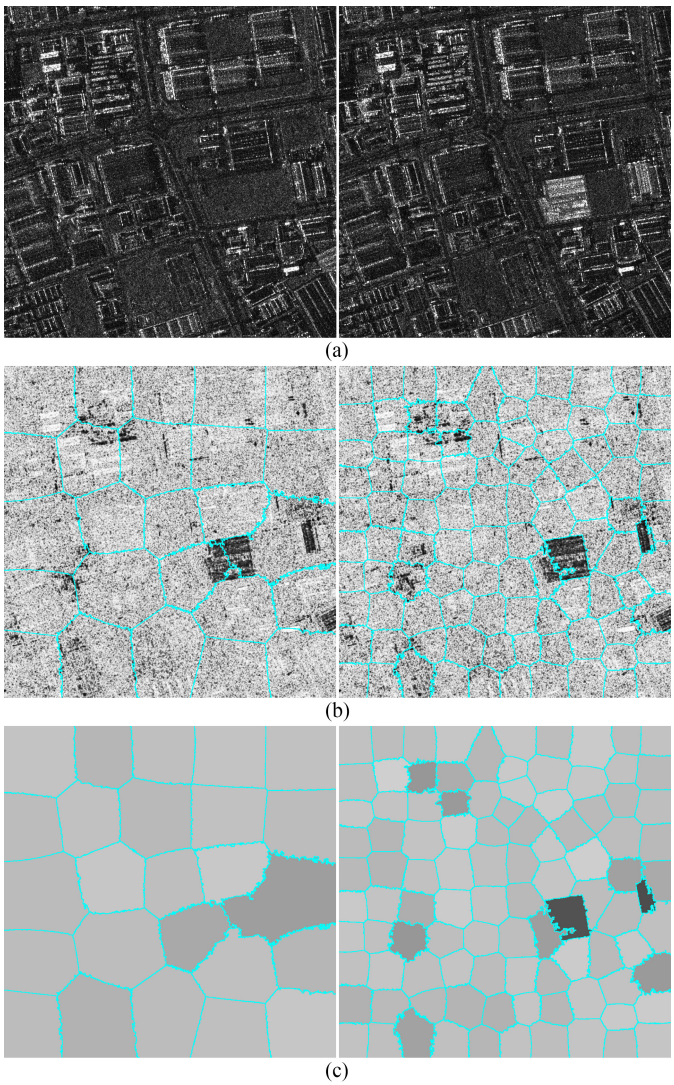


Fig. 6. Examples of generating the R image using the real SAR image pair: (a) extracted images from dataset 3 with the size of 800×800; (b) Segmented results of SLIC overlaying on the MR image with the initial number of superpixels $k = 30$ and $k = 100$, respectively; and (c) R images estimated by the proposed method corresponding to the segmentation results in (b).

It is observed that the value of R is closely related to the number of superpixels, thus making the setting of the number of superpixels an important issue needed to be discussed. And the problem will be specifically discussed in Section III-C.

III. Experiments and Analysis

A. Available Image Pairs

Three SAR image pairs are selected for the analysis of the experimental results. Among them, datasets 1 and 2 are utilized to evaluate the effectiveness and performance of the proposed estimator. In this section, the new estimator is compared with the other common change statistics and classic thresholding methods, respectively. In addition, datasets 2 and 3 are used for the setting test of the superpixels number. Due to the characteristic of a large dynamic range [26], the SAR images used in this article are 16 bits in order to keep more details of radar echo. The available sets of image pairs and their detailed information are shown in Table II.

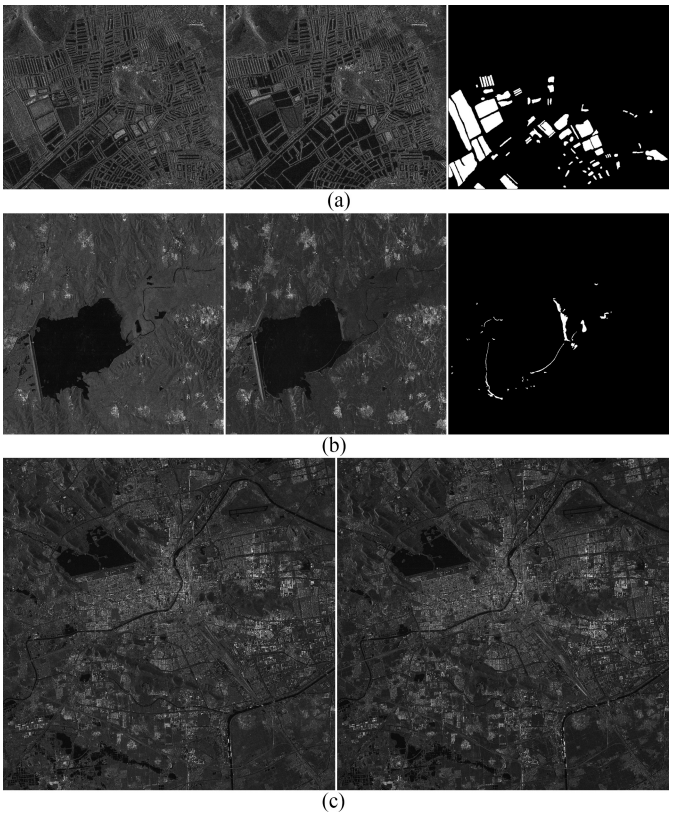


Fig. 7. Experimental image pairs: (a) dataset 1 and ground truth; (b) dataset 2 and ground truth; and (c) dataset 3.

Since performing a quantitative comparison of the proposed method and other four typical thresholding approaches using datasets 1 and 2, we give the image pairs and the corresponding ground truth in Fig. 7(a) and (b), respectively. The ground truth is determined by visual inspection and recorded as a binary mask, as shown in the right figures of Fig. 7(a) and (b). Dataset 1 mainly covers aquaculture bases, the change of which is our interest. Dataset 2 is utilized to identify the icing areas in the reservoir by using the unfrozen image in October 2015 and the iced image of the following year. Dataset 3 covers the areas of Xuzhou City and its suburbs, containing various surface features such as water, building, and vegetation [Fig. 7(c)]. Choosing these images would be more convincing to illustrate the influence of the superpixels number on the detection result than images with simple coverage.

B. Evaluation Criteria

The following criteria are used to evaluate the performance of the proposed estimator.

Precision of Change (Pc): the ratio of the correctly detected change pixels to all the detected change ones.

Recall Ratio (R): also known as the positive rate which is the ratio of the detected change pixels to all the true change ones.

Overall Accuracy (OA): the ratio of all correctly detected pixels in the change areas and the background to all pixels of the image.

Kappa Coefficient [27]: a comprehensive evaluation measure taking both the correct and false detected pixels into

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consideration can be calculated from the confusion matrix. The higher the value is, the more consistent the true classification image and the predicted image is.

To fully evaluate the effectiveness of methods, we need to take the change precision and the recall ratio into consideration at the same time. Ideally, the higher the two indicators are, the better the algorithms are. Unfortunately, the two criteria are always opposed to each other, which means that the increasing precision usually reduces the recall rate. Therefore, we take the two comprehensive measures, OA and Kappa coefficient, into account as well.

C. Results and Analysis

This section is divided into three parts. First, we discuss how to set the initial superpixels number of SLIC depending on the requirement of applications. Then, the new estimator is compared with the change statistics commonly used in coherent and noncoherent methods in terms of initial detection results. Finally, the performance of the new change statistic is evaluated by comparing it with classical thresholding algorithms. The proposed method is developed with MATLAB 2019.

It is worth mentioning that there are different names for the detection images obtained by the change statistics in coherent and noncoherent algorithms. Specifically, in coherent methods, raw estimator outputs are often called SAR CD images [3], whereas images obtained using intensity-based statistics, such as the subtraction operator, are usually called DIs in noncoherent approaches. For consistency with the definition of raw estimator outputs, DI is referred to as the SAR CD image as well in this article. After thresholding these images, the obtained results are called CD maps.

1) Experiment of Setting Superpixels Number: In this section, we performed a comparison of SAR CD images for different superpixel numbers. Fig. 8(a) demonstrates the CD images obtained using the proposed estimator for different initial superpixels numbers of $k = 1$, $k = 100$, $k = 200$, and $k = 300$ in dataset 3. Obviously, $k = 1$ means directly getting the detection image using the proposed estimator without any segmentation.

Before showing the results, we should mention that there are strong and weak types in change. For example, the appearance or disappearance of the buildings between two data collections should be defined as the strong change. In cases like that, the intensity difference between the two images would be glaringly obvious as the buildings always present very bright in SAR images. But for low vegetation, especially for crops, they appear relatively dark in SAR images, thus making the intensity difference due to growth during two collection times not significant as buildings. This change can be defined as the weak one. According to the requirements of the specific applications, the weak changes sometimes need to be detected, for instance, in some agricultural tasks, but many times need to be ignored because the growth of vegetation is inevitable during two acquisition times, thus avoiding confusion with changes we care about.

To detailedly and intuitively show the difference, parts of Fig. 8(a), corresponding to the areas marked in red and yellow rectangles, are zoomed in and shown in Fig. 8(b) and (c), respectively. Their corresponding SAR images extracted from dataset 3 are shown in Fig. 9. It is observed that the red-rectangle area is mainly covered with crops [Fig. 9(a)], but the yellow rectangle is the building area [Fig. 9(b)]. Therefore, the changes in these two areas correspond to the weak and strong changes, respectively. Note from Fig. 8(b) that the extracted change in detail can be increased with the superpixels number increasing, which means that the extraction of weak change is closely related to the superpixels number. However, in Fig. 8(c), the extracted changes almost keep the same with each other, thus indicating the little influence of the superpixels number on the strong CD. Therefore, the introduction of SLIC makes the new estimator adjustable to extract the changes at different degrees. Specifically, the more the number of superpixels is set, the more details of the weak change are extracted, which makes it more convenient for us to detect change according to the specific requirements of tasks.

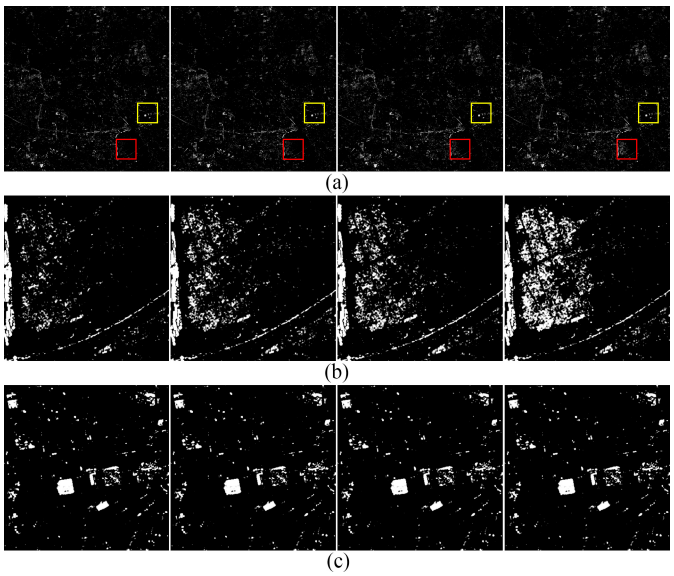


Fig. 8. SAR CD images for different numbers of superpixels $k = 1$, $k = 100$, $k = 200$, and $k = 300$ in: (a) dataset 3 and corresponding zoomed-in view images of the: (b) red rectangle and (c) yellow rectangle.



Fig. 9. SAR image pairs corresponding to the: (a) red rectangle and (b) yellow rectangle extracted from dataset 3.

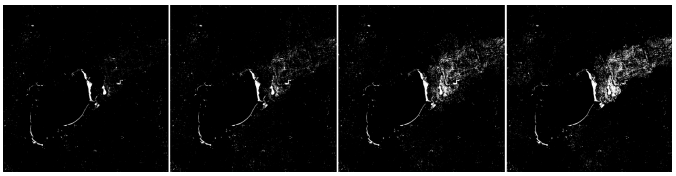


Fig. 10. SAR CD images for different superpixels numbers of $k = 1$, $k = 10$, $k = 30$, and $k = 50$ in dataset 2.

To further validate the adjustability of the proposed method, we also experiment in dataset 2. Fig. 10 gives the CD results

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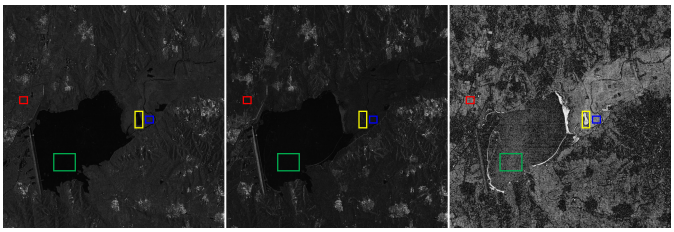


Fig. 11. SAR image pair of dataset 2 and its MR image.

for the superpixels number of $k = 1$, $k = 10$, $k = 30$, and $k = 50$. For dataset 2, we aim to detect the added ice area, for example, the yellow-rectangular area in Fig. 11. Note that the MR image fills in these ice-changed regions with significant bright pixels, as shown in the right figure in Fig. 11. However, the weak change caused by the vegetation growth, such as the area marked in the red rectangle, brings trouble to us. The difference between the two images in these areas is also enhanced in the MR image, appearing as bright pixels as well. If the threshold is not set properly, the weak change in these areas could be easily extracted. Considering the conclusion we have demonstrated in dataset 3, we set the number of the superpixels to the small values to ignore these weak changes. And the results in Fig. 10 demonstrate that the changes caused by this vegetation growth are almost not extracted in the condition of $k < 50$.

Moreover, we find another advantage of the proposed method; the change between the strong and weak one can be similarly extracted at different degrees by setting the number of superpixels. As can be seen from the MR image, the brightness of the area marked in the blue rectangle, i.e., the area along the river, is slightly brighter than the red-rectangle area but darker than the yellow-rectangle area. Thus, the change along the river basin is somewhere between the vegetation and the ice. And the results in Fig. 10 shows that the detected area along the river also gradually increases with the number of superpixels, which further demonstrate the adjustability of new statistic to extract different degree changes. Consequently, if trying to extract more detailed change information containing the weak change, one can set a larger number of superpixels and, conversely, a smaller one.

The introduction of SLIC can help us adapt the proposed method to detect changes at different degrees according to the requirements of tasks. One point to be mentioned here is that all the detected changes are the initial SAR CD images without any postclassification processing.

2) Comparison With Other Change Statistics: As a combination of coherent and noncoherent algorithms in a theoretical derivation way, the proposed estimator is compared with other five change statistics commonly used in these two kinds of methods: MR operator [5], log-ratio (LR) operator [12], neighborhood-based ratio (NR) operator [6], typical coherence estimator [19], and alternative coherence estimator proposed by Seymour and Cumming [20]. Among them, the first three are change statistics of intensity-based methods, and the remaining two are that of coherent methods. Fig. 12 shows SAR CD images obtained with these different change statistics in datasets 1 and 2, respectively. In this experiment, the spatial window size to generate detection images by the MR and NR operators is set to 4×4. Because the multilooking parameter is set to 3×3, the window size to calculate coherence magnitude using the initial single look complex (SLC) images is also set to 3×3 for coherent algorithms, thus keeping the size of the detection images the same as the multilooking images. For the proposed method, the initial numbers of superpixels for the two image pairs are set to 30 and 1, respectively.

It is observed from Fig. 12 that the proposed estimator can yield higher-contrast SAR CD images than other change statistics, favorable for the subsequent thresholding. For three intensity-based change statistics, SAR CD images have a relatively obvious contrast between the changed and unchanged areas in dataset 1. The change areas are filled with bright pixels, whereas the unchanged areas are with dark pixels. But for coherent methods, the results show the opposite. In addition, note that the detection image obtained by the LR operator presents a lower brightness than that of MR and NR operators. The above conclusion can also be proved by the results from dataset 2. As we know, these intensity-based methods are very sensitive to intensity changes. For a pixel in the SAR image pair, it would appear as a bright pixel in the SAR CD image if there is an intensity difference in the local neighborhood of the pixel. Therefore, the detection images of these methods in Fig. 12 are filled with brighter pixels in the change areas of the aquaculture type in dataset 1 and the reservoir frozen areas in dataset 2. However, it is precisely the sensitivity to intensity changes that makes these intensity-based methods have an obvious weakness. Specifically, the weak intensity change between the two images of the same local neighborhood would be magnified on the detection images with these methods, as shown in the red-rectangular areas in Fig. 12(a). However, it can be seen from the detection images of coherence estimators in Fig. 12(a) that changes are not obvious in the red-rectangle areas. Coherence estimators are less sensitive to intensity changes than intensity-based change statistics. These magnified weak changes in the SAR detection image obtained by the above intensity-based operators will directly affect the precision of the final change map. Although essentially an intensity-based approach, the proposed estimator combines the advantage of coherent methods to some extent, not as sensitive to intensity changes as the other three intensity-based change statistics in these areas.

Although datasets 1 and 2 are repeat-pass SAR image pairs, the decorrelation between each pair is very serious, as can be seen from the coherence magnitude images obtained by the typical coherence estimator. There are several reasons for this decorrelation. First, temporal decorrelation is inevitable due to different acquisition times. In addition, decorrelation or low coherence due to the low cluster-to-noise ratio (CNR) of water and the volume scattering of vegetation is also obvious in the two datasets. In addition to the above sources of decorrelation, change is also an important factor of decorrelation. Therefore, the purpose of the coherence estimator is to distinguish change-decorrelation areas from the low-coherence areas. Note that the detection image obtained by the typical coherence estimator fails to achieve this purpose, which is further demon-

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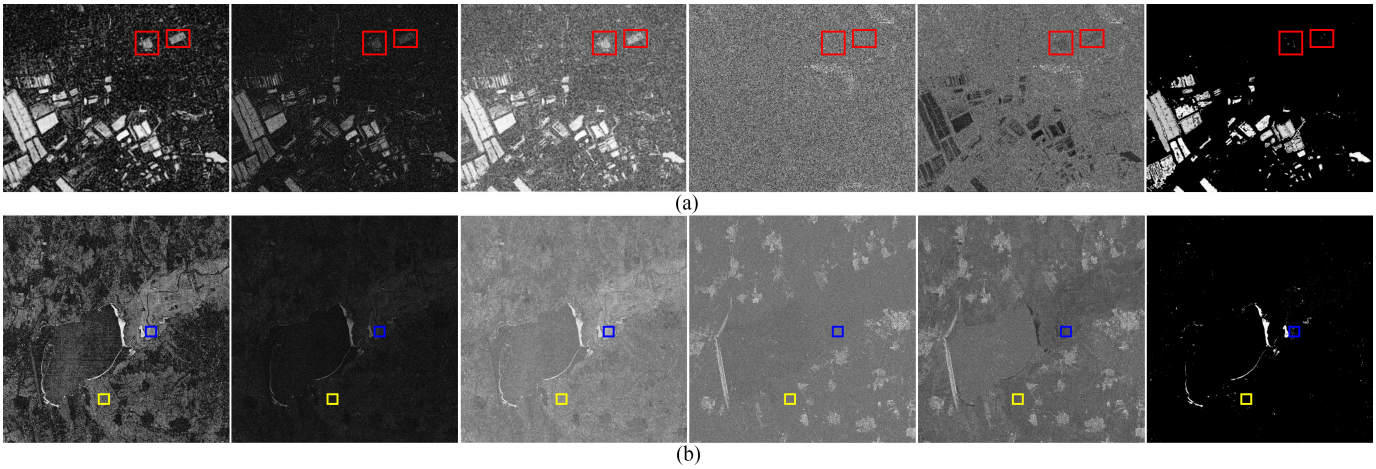


Fig. 12. SAR CD images generated with different change statistics in: (a) dataset 1 and (b) dataset 2. The detection images from left to right correspond to the results obtained by the MR operator, LR operator, NR operator, typical coherence estimator, alternative coherence estimator, and the proposed estimator, respectively.

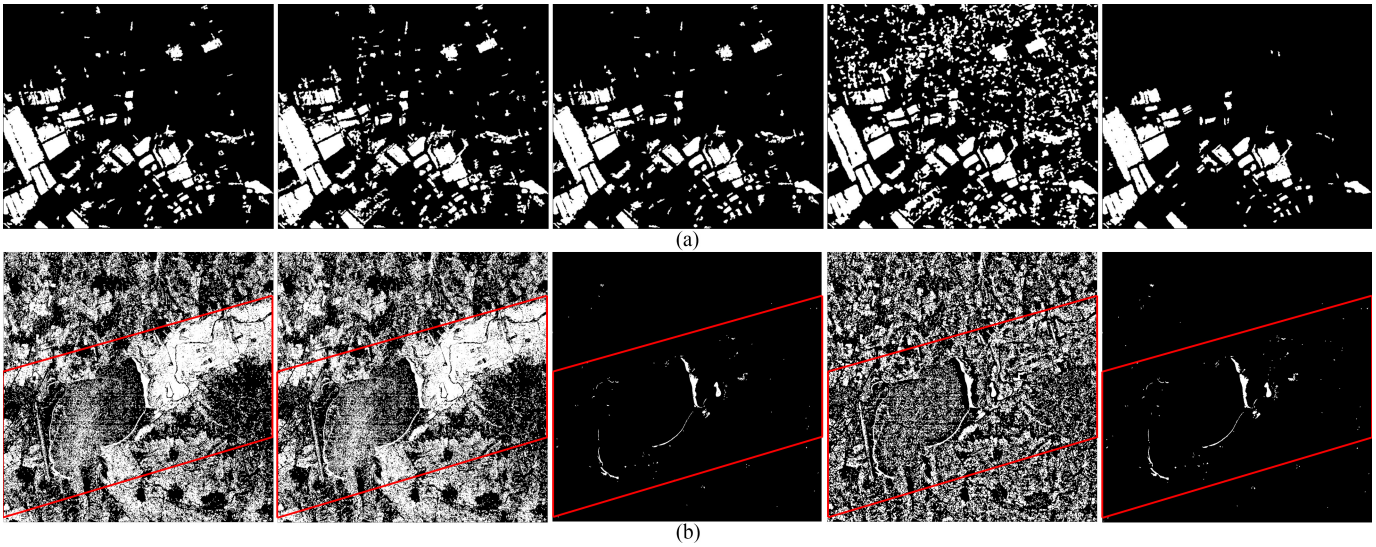


Fig. 13. CD maps of Otsu, IT, ME, AT, and the proposed method in: (a) dataset 1 and (b) dataset 2.

strated by the result shown in Fig. 12(b). By contrast, the alternative coherence estimator is more suitable for SAR CD, filling change areas with darker pixels compared with low-CNR and volume-scattering areas. However, the estimator provides low coherence values in all decorrelation regions, thus resulting in a reduced contrast between the change and other decorrelation regions, especially for dataset 2. As described in Section III-C1, for dataset 2, the intensity change in areas along the river marked in the blue rectangle is slightly stronger than that of farmland areas marked in the yellow rectangle. Similarly, the coherence magnitude along the river basin is slightly lower than that in farmlands [see the fifth figure in Fig. 12(b)]. Obviously, these two weak changes will bring trouble to the subsequent thresholding processing. Attractively, the new estimator can effectively filter these two types of changes by setting a small initial number of superpixels, as shown in the right figure in Fig. 12(b). In addition, it can also extract the areas along the river by setting a large number of superpixels, as shown in Fig. 10. Although the new estimator is designed to combine coherent and noncoherent methods in a theoretical derivation way, its results also combine the advantages of these two types of methods to some extent, thus effectively highlighting the change areas and filtering the weak-change areas.

3) Performance of the Proposed Estimator: In this section, we compare the CD map of the proposed approach with four classic thresholding segmentation methods: Otsu [28], iteration threshold (IT) [29], max entropy (ME) [30], and adaptive threshold (AT) in OpenCV. In the test, only one needed parameter, the spatial window size to estimate the MR image, is set to 4×4 except for the AT method. For the AT method, we also need to set the block-size parameter as 105 to calculate the weighted sum and the tolerance as -7 to reduce the effect of the speckle noise. The CD maps of these methods in datasets 1 and 2 are shown in Fig. 13, and their corresponding quantitative precision results are demonstrated in Table IV. As the tasks do not need to extract the weak change caused by the vegetation growth, we set small number values of superpixels for the two image pairs: 30 for dataset 1 and 1 for dataset 2. For dataset 2, we aim to detect the added ice area

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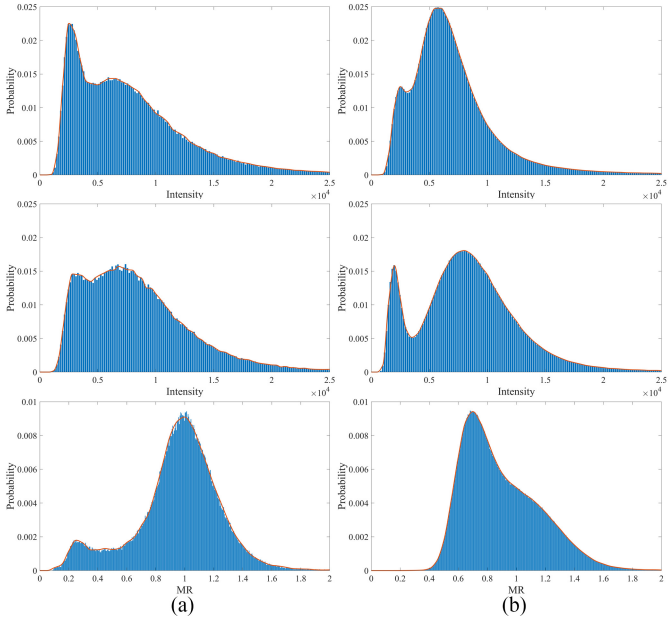


Fig. 14. Statistical histograms of the SAR image pair and its corresponding MR image: (a) dataset 1 and (b) dataset 2.

and ignore the weak changes in areas of the farmland and along the river, as mentioned Section III-C1, so the smaller the number of superpixels is set, the better the detection result is. In addition, to filter the small blocks and reduce the influence of the speckles, a brief sieve process is performed for the initial detection maps, and the sieve threshold is set to 20 pixels, which means that the size of the blocks is less than the value would be ignored.

Before analyzing the CD results, we analyzed the inherent properties of the datasets and introduced the thresholding method for the proposed estimator. The statistical histograms of two image pairs and the corresponding MR-image histograms are shown in Fig. 14. For image pairs used in this article with a pixel bit of 16, the grayscale range is from 0 to 65 535. Here, the bin number of histograms of all images is set to 400 for convenience of comparison. In Fig. 14, the x-axis corresponds to the intensity or value range of the image and the y-axis "Probability" is the ratio of the number of pixels in each bin to the total pixel number of the whole image. And the red line is a simple fitting curve of the histogram.

Ideally, the statistic histograms of two SAR images with the same collection conditions should keep nearly the same trend, assuming that no change occurs in the area covered by the two images between collection times. However, due to the ground-feature changes during the collections and different collection conditions such as the atmosphere, they always have differences in various degrees in reality. It is observed from the upper two figures in Fig. 14(a) that the two histograms estimated from dataset 1 almost keep the same trend. In addition, the two peaks of the histograms also have approximately the same intensity values. Specifically, it can be seen with the help of the two red fit curves that the two peak values of one image correspond to about 2500 and 6500, and the others are about 2900 and 6800. Different from dataset 1, there are significant differences in the trend of the histograms for dataset 2 [see the top two figures of Fig. 14(b)]. The peak intensity values of the top image are about 2500 and 5800, whereas the middle image corresponds to about 2000 and 8000. Therefore, the statistical characteristics of the two datasets in intensity are different from each other, which can be further demonstrated by their MR images. For dataset 1, note that the histogram of its MR images has two obvious peaks [bottom figure in Fig. 14(a)]. Among them, the larger one represents the background (unchanged area). In these areas, the mean value of the MR image is about 1, which means that the mean intensity of the two images is similar to each other in a neighborhood of one pixel. However, the larger peak value is about 0.7 for dataset 2. Thus, these two datasets have different statistical characteristics, thus making the applicability of the new method more convincing by taking them as experimental data.

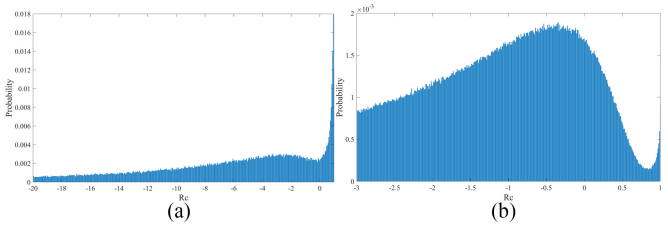


Fig. 15. Statistical histograms of R_c images: (a) dataset 1 and (b) dataset 2.

As previously mentioned, the output of the new estimator represents the detection result before thresholding. To get the final change map, it is necessary to apply a decision threshold to the histogram of the CD image estimated by the new estimator. Fig. 15 gives the statistical histograms of R_c images corresponding to datasets 1 and 2, respectively. Due to the stretching effect of the new estimator, the histograms of R_c images have two significant peaks, thus making it possible to get the decision threshold with a commonly used thresholding method such as Otsu. According to the threshold, the final change map of the proposed method can be easily obtained.

By analyzing the CD maps of different methods, the following conclusions can be drawn.

For the image pair with similar statistical characteristics, having two significant peaks for the histogram of its MR image, such as dataset 1, all methods except AT are applicable, but the proposed method can obviously improve the detection precision.

For the image pair with different statistical characteristics, without two peaks for the histogram of its MR image, such as dataset 2, ME and the novel estimator achieve better results whereas the other three methods are not suitable anymore.

The specific analysis of the above conclusions is as follows.

The histogram of the MR image of dataset 1 has two obvious peaks where the high one stands for the background and the small one for the change, which means that the change area and the background of the dataset are relatively easy to identify. Thus, all methods except AT are suitable for dataset 1. However, it can be known from the measurement Pc in Table III that the other four methods detect many false alarms. As we know, if the statistical histogram of an image has two significant peaks and the probability corresponding to the valley is close to zero, the changed and unchanged

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TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS

Set	method	Pc	R	OA	Kappa
1	Otsu	0.6928	0.9627	0.9524	0.7794
	IT	0.5742	0.9799	0.9234	0.6831
	ME	0.7101	0.9585	0.9556	0.7912
	AT	0.3646	0.9416	0.8257	0.4433
	proposed	0.8823	0.9188	0.9791	0.8885
2	Otsu	0.0267	0.9974	0.5372	0.0279
	IT	0.0257	0.9975	0.5199	0.0260
	ME	0.9279	0.8235	0.9969	0.8710
	AT	0.0347	0.9936	0.6488	0.0436
	proposed	0.8770	0.8886	0.9970	0.8812

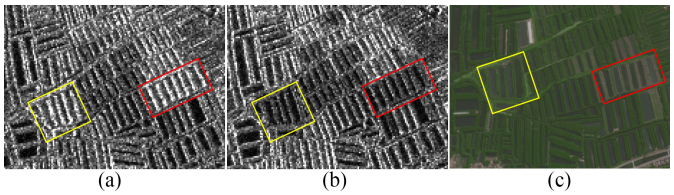


Fig. 16. Areas of the false alarm in dataset 1: (a) and (b) close-up views of the image pair, and (c) corresponding optical image captured from the Baidu map.

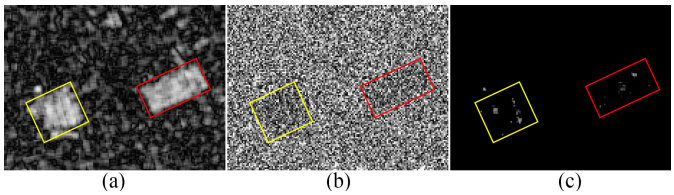


Fig. 17. SAR CD images of Fig. 16(a) and (b) generated from (a) MR operator, (b) alternative coherence estimator, and the (c) proposed method.

areas would be easily identified. But for dataset 1, the valley's probability is obviously not zero, as shown in the bottom-left figure in Fig. 14, which means that the boundary to identify the change and the no-change is not so clear, thus resulting in many false alarms.

Visibly different from dataset 1, the MR-image histogram of dataset 2 has no apparent two peaks, thus making it difficult to detect change information for the dataset. Therefore, the Otsu, IT, and AT methods are no longer suitable for CD of dataset 2. Though having a high detection precision (0.9279), ME has an unsatisfactory point with a lower positive rate (0.8235), thus indicating that many changed pixels have not been extracted. The proposed strategy comparatively increases the recall ratio but decreases the change precision, so it seems more like a compromise between Pc and R. However, the new method is slightly better than ME from the comprehensive evaluation measures OA and Kappa coefficient.

Moreover, we make a detailed analysis of the areas where false alarm always occurs. For dataset 1, the false alarm mainly occurs in areas of weak changes caused by vegetation growth, for example, the regions marked in red and yellow in Fig. 16(a) and (b). Although there are some differences in intensity in these areas [see Fig. 17(a)], the ground features have not changed in actuality, which can be further demonstrated by Fig. 17(b) where the coherence values in these marked areas are not obviously different from other areas. And the corresponding optical image captured from Baidu Map clearly shows that these areas of false alarm are covered



Fig. 18. CD maps of Otsu, IT, ME, AT, and the proposed method in images of Fig. 16(a) and (b).

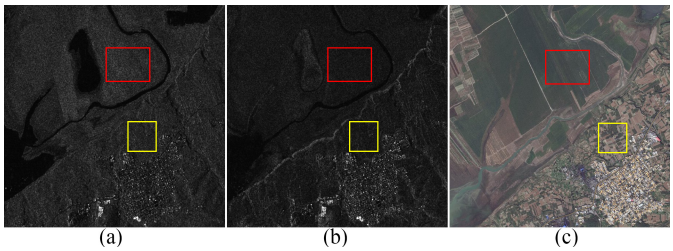


Fig. 19. Areas of the false alarm in dataset 2: (a) and (b) close-up views of the image pair, and (c) corresponding optical image captured from Esri ArcGIS.

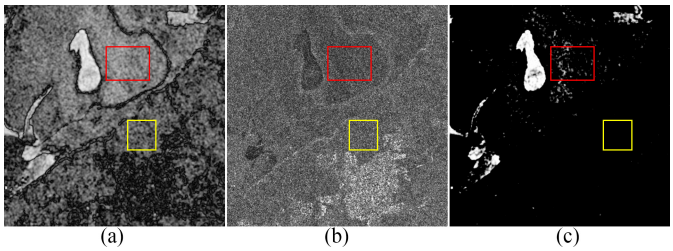


Fig. 20. SAR CD images of Fig. 19(a) and (b) generated from (a) MR operator, (b) alternative coherence estimator, and the (c) proposed method.

with green vegetation. However, these intensity differences due to vegetation growth are not what we want to extract. For the two extracted images, their corresponding CD maps of different methods are shown in Fig. 18. It is clearly seen that the proposed statistic can greatly filter these differences by setting a small number of superpixels, as demonstrated in Section III-C1, thus consequently improving the detection performance.

For dataset 2, there are two main reasons for many methods present many false alarms: one is the significant difference of the mean backscatter power between the image pair and the other is various types of change. It is well known that smooth water surfaces often appear as dark pixels in SAR images due to low CNR. With no changes happening during two data acquisitions and no obvious differences existing in the mean power of two SAR images, the values estimated by the MR operator [5] in water areas should be around 0.0, shown in black on the MR image. However, it is observed from the green-rectangle region of the MR images in Fig. 11 that the MR values in water areas are obviously not 0.0, thus making it difficult to get good detection results. This power difference can be reduced to a certain extent by some preprocessing such as radiometric calibration.

In addition to the difference in the mean backscatter power, the weak change at different degrees is the other important reason for the false alarms. Same as dataset 1, the false alarm also mainly occurs in weak-change areas caused by vegetation growth, for example, the rectangular areas marked in red and yellow in Fig. 19(a) and (b). Note from Fig. 19(c) that these areas are farmland; however, the crop growth is different.

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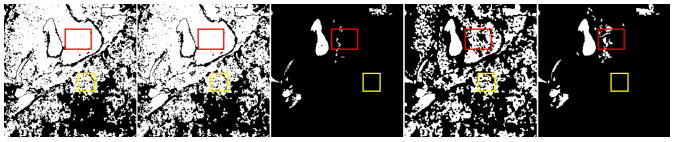


Fig. 21. CD maps of Otsu, IT, ME, AT, and the proposed method in images of Fig. 19(a) and (b).

In addition, it is also observed from Fig. 20(a) and (b) that the MR values and coherence magnitudes of these two marked areas are different, thus indicating different degrees of change. Attractively, the proposed estimator achieves the same performance in dataset 2 as in dataset 1 by setting a small number of superpixels, thus effectively filtering the weaker change caused by vegetation growth [see the yellow-rectangular area in Fig. 20(c)]. Furthermore, the detected area along the river can increase with the number of superpixels, as shown in Fig. 10. Fig. 21 shows the corresponding CD maps of different methods. Note that Otsu, IT, and AT extract most areas of these two types of weak changes, unfavorable for identifying ice-change areas, whereas ME and the new methods get comparatively better performance. However, many changed pixels are not extracted by ME, for example, hollow holes in the ice-change area in the middle-top part of the ME's change map in Fig. 21. Although extracting some weak-change areas, the proposed method can detect most areas of the ice change, thus giving a higher value of recall ratio than ME. One point to be mentioned here is that the quantitative results of dataset 2 in Table III are calculated in the region marked in red [see Fig. 13(b)] as the task mainly aims to detect changes in the water area.

To sum up, the proposed estimator almost gives the best performance of CD. The statistical histograms of R_c images in Fig. 15 are a realistic explanation of the conclusion. The new approach has a good stretching effect on the histogram of the MR image, which is consistent with the conclusion demonstrated in Section II-B that the new estimator can largely stretch the values of the change and background to the opposite directions. This stretch effect is more obvious in dataset 2 as shown in Fig. 15(b). Therefore, it is concluded that the novel estimator has broad applicability to SAR image pairs with different statistical characteristics. And the proposed CD method is adaptive and adjustable for SAR images to detect changes. The reasons are as follows.

1) The proposed method can automatically determine the stretched threshold and stretch the histogram of the MR image to the one with two obvious peaks.
2) The introduction of SLIC can help to detect changes at different degrees by setting different numbers of the initial superpixels according to the specific requirements of tasks.

IV. Conclusion

Inspired by the derivation of the coherence estimators in CCD, we proposed a new ML intensity-based estimator to combine the advantage of intensity-based detection methods and CCD algorithms in a theoretical way. First, the estimator is theoretically proved as an adaptive thresholding method, which can automatically determine a threshold for each pixel and then positively or negatively stretch the intensity value with reference to the threshold. Then, the introduction of SLIC into the calculation of the estimator makes it adjustable to detect changes at different degrees, which is further demonstrated in real SAR images. Finally, the experimental results show that the new estimator can combine the advantages of coherent and noncoherent algorithms to some extent and obtain better CD maps than other classic thresholding methods in SAR image pairs with different statistical characteristics. In future work, we will evaluate the effectiveness of the proposed estimator in more SAR images.

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